

Project Title: NeuroVision - Brain Tumor Classification

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Abstract

This project, titled NeuroVision, presents an automated system designed for brain tumor classification using Magnetic Resonance Imaging (MRI) scans. Traditional manual diagnostics can be time-consuming and prone to error, necessitating efficient computational alternatives. We propose a custom, lightweight Convolutional Neural Network (CNN) optimized for binary classification to distinguish between normal and tumor-afflicted brain images. The dataset, comprising 253 MRI scans, underwent rigorous preprocessing, including grayscale conversion, standardizing resolution to 128x128 pixels, and min-max pixel normalization. The neural network architecture utilizes sequential Conv2D and MaxPooling2D layers for automated feature extraction, coupled with a Dropout layer to actively mitigate overfitting. Trained using the Adam optimizer and binary crossentropy loss, the model achieved a robust validation accuracy of approximately 82%. The results demonstrate that a streamlined CNN balances high recall with computational efficiency, making it highly suitable for deployment as a rapid second-opinion diagnostic tool.

1. Introduction

The application of Deep Learning, particularly Convolutional Neural Networks (CNNs), has significantly transformed the field of medical image analysis. The primary objective of the NeuroVision project is to build an automated diagnostic tool that classifies brain MRI scans into 'Tumor' (Yes) or 'Normal' (No) categories.

While highly complex, pre-trained models like VGG16 or ResNet50 are available through transfer learning, they are computationally heavy and typically require massive datasets to fine-tune effectively. The core problem this project addresses is the need for a well-optimized, lightweight custom CNN that can achieve comparable diagnostic accuracy with significantly less computational overhead on smaller datasets. Ultimately, NeuroVision aims to balance computational efficiency with high recall (minimizing dangerous false negatives) to serve as a reliable second-opinion tool for medical professionals.

2. Literature Review

The architectural design and preprocessing pipeline of NeuroVision were heavily influenced by key findings in recent medical AI research.

- **Automated Feature Extraction:** Traditional machine learning approaches required time-consuming manual feature extraction, such as GLCM features. Foundational research demonstrated that CNNs successfully automate this process, extracting hierarchical features like edges and tumor masses directly from raw MRI pixels. Research also proved that utilizing the ReLU activation function drastically speeds up hidden layer training compared to older functions like Tanh or Sigmoid.
- **Impact of Normalization:** Medical images often suffer from varying contrast levels. Studies showed that applying Min-Max Normalization to scale pixel intensities to a 0–1 range prevents the exploding gradient problem and allows the Adam optimizer to converge much faster.
- **Handling Overfitting:** A major challenge in medical image classification is the limited availability of annotated datasets, which frequently leads to model overfitting. Researchers concluded that introducing a Dropout layer, which randomly deactivates a percentage of neurons during training, forces the network to learn robust, generalized features.

3. Methodology & Data Preparation

Proper data preparation is crucial to ensure the neural network receives consistent and mathematical inputs.

3.1 Dataset Details

- **Source:** The dataset consists of 253 human brain MRI scans sourced from open-source repositories.
- **Classes:** It is structured as a binary classification problem. Class 0 represents healthy brains without tumors, while Class 1 represents brains with distinct tumor regions or structural deformations.

3.2 Preprocessing Pipeline

- **Color Conversion:** All raw images were converted to Grayscale to reduce computational complexity while retaining necessary structural features.
- **Resizing:** Every scan was uniformly resized to a standard 128x128 pixel resolution to ensure consistent input dimensions for the neural network.
- **Normalization:** Raw pixel intensity values, which range from 0 to 255, were divided by 255.0 to scale them down to a 0.0–1.0 range. This scaling helps the neural network learn faster.
- **Train-Test Split:** The dataset was randomly shuffled and divided using an 80:20 ratio. Exactly 202 images were allocated for training the model, while 51 images were held back as unseen testing data to validate performance.

4. System Architecture

The project utilizes a Sequential Convolutional Neural Network built using the Keras API, processed layer by layer.

- **Convolutional Blocks (Conv2D):** The architecture incorporates multiple Conv2D layers acting as feature extractors. The model uses three sequential blocks with 32, 64, and 128 filters to identify hierarchical patterns ranging from simple edges to complex tumor textures.
- **Spatial Downsampling (MaxPooling2D):** MaxPooling layers are utilized after convolutional layers to reduce the spatial dimensions of the image by half, which compresses the data and helps prevent overfitting.
- **Flatten Layer:** A Flatten layer is used to transition the 2D feature maps generated by the convolutional blocks into a single 1D numerical array.
- **Fully Connected (Dense) Layers:** The flattened data is passed to a Dense layer with 128 neurons acting as the model's "brain" to interpret the combined features. The final output layer consists of a single neuron with a Sigmoid activation function to squash the output into a probability score between 0 and 1 for binary classification.
- **Regularization (Dropout):** A Dropout layer with a rate of 0.5 was integrated to randomly turn off 50% of the neurons during training. This proved critical in maintaining a viable validation accuracy of ~82% despite the small dataset size.

5. Implementation & Results

The model's learning process was configured to minimize errors and actively monitor its predictive performance.

5.1 Training Configuration

- **Optimizer and Loss:** The model utilized the highly efficient Adam optimizer to adjust the neural network's weights. Because it is a binary classification problem, 'binary crossentropy' was tracked as the standard loss function to measure prediction errors.
- **Training Loop:** The model was trained over 20 epochs, processing the training images in mini-batches of 16 to update its mathematical weights efficiently.

5.2 Performance Evaluation

- **Classification Report:** The model's performance was evaluated using standard metrics, including Accuracy (overall correct predictions), Precision, Recall (Sensitivity), and the F1-Score. Achieving high Recall was prioritized to ensure actual tumors were successfully identified, minimizing misdiagnoses.
- **Confusion Matrix:** A color-coded heatmap was generated to visually interpret the model's classifications. This allowed for a direct visual breakdown of True Positives, True Negatives, False Positives, and False Negatives.

6. Conclusion

The NeuroVision project successfully demonstrates that a well-optimized, lightweight custom CNN can efficiently classify brain MRI scans. By implementing a rigorous preprocessing pipeline and leveraging a sequential architecture with targeted regularization (Dropout), the model achieved a validation accuracy of approximately 82%. This approach minimizes computational overhead while maintaining the high recall necessary for medical applications, validating its potential as a rapid diagnostic tool.

Future Scope: Moving forward, the natural progression for this research is to deploy the trained model as a functional web application. This would provide an accessible interface for end-users and medical practitioners to upload raw MRI scans and receive real-time, automated classification predictions directly in their browsers.